Subquery

Baza danych: PostgreSQL

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Subquery – one row

	first_name	last_name	department_id
1	Matthew	Weiss	50
2	Adam	Fripp	50
3	Payam	Kaufling	50
4	Shanta	Vollman	50
5	Kevin	Mourgos	50
6	Julia	Nayer	50
7	Irene	Mikkilineni	50
8	James	Landry	50

```
select
first_name,
last_name,
department_id
from employees
where
department_id =
(select department_id from departments where department_name
= 'Shipping')
```

Subquery – one row - error

```
SQL Error [21000]: BŁĄD: ponad jeden wiersz zwrócony przez podzapytanie użyte jako wyrażenie
```

```sql
select
    first_name,
    last_name,
    department_id
from employees
where
department_id =
(select department_id from departments
       where department_name in ('Shipping','Administration'))
```

Subquery – more than one row

	first_name	last_name	department_id
1	Matthew	Weiss	50
2	Adam	Fripp	50
3	Payam	Kaufling	50
4	Shanta	Vollman	50
5	Kevin	Mourgos	50
6	Julia	Nayer	50
7	Irene	Mikkilineni	50

```
select
    first_name,
    last_name,
    department_id
from employees
where
department_id IN
        (select department_id from departments
                where department_name in
                        ('Shipping','Administration'))
```

Subquery – many in one query

	first_name	last_name	department_id
1	David	Williams	60
2	Valli	Jackson	60
3	Diana	Nguyen	60
4	Nandita	Sarchand	50
5	Alexis	Bull	50
6	Sarah	Bell	50
7	Jennifer	Whalen	10

```
select
  first_name,
  last_name,
  department_id
from employees
where salary between (select min_salary from jobs
                      where job_title = 'Programmer')
                 and (select max_salary from jobs
                      where job_title = 'Stock Clerk')
```

Subquery – in having

	department_id	sum
1	70	10,000
2	[NULL]	7,000
3	10	4,400
4	40	6,500

```
select
    department_id,
    sum(salary)
from employees
group by department_id
having sum(salary) < (select sum(salary) from employees
                      where department_id = 20)
```

Subquery – operator ANY

	first_name	last_name	salary	department_id
1	Steven	King	24,000	90
2	Neena	Yang	17,000	90
3	Lex	Garcia	17,000	90
4	Alexander	James	9,000	60
5	Nancy	Gruenberg	12,008	100
6	Daniel	Faviet	9,000	100
7	John	Chen	8,200	100

```
select
first_name,
last_name,
salary,
department_id
from employees
where salary > ANY (select salary from employees
                    where department_id = 20)
```

Subquery – operator ALL

	first_name	last_name	salary	department_id
1	Steven	King	24,000	90
2	Neena	Yang	17,000	90
3	Lex	Garcia	17,000	90
4	John	Singh	14,000	80
5	Karen	Partners	13,500	80

```
select
    first_name,
    last_name,
    salary,
    department_id
from employees
where salary > ALL (select salary from employees
                    where department_id = 20)
```

Subquery – correlation

	first_name	last_name	salary	department_id
1	Steven	King	24,000	90
2	Alexander	James	9,000	60
3	Bruce	Miller	6,000	60
4	Nancy	Gruenberg	12,008	100
5	Daniel	Faviet	9,000	100
6	Den	Li	11,000	30
7	Matthew	Weiss	8,000	50

```
select
first_name,
last_name,
salary,
department_id
from employees e
where salary > (select avg(salary) from employees
                where department_id = e.department_id)
```

Subquery – correlation - EXISTS

	department_name
1	Administration
2	Marketing
3	Purchasing
4	Human Resources
5	Shipping
6	IT
7	Public Relations
8	Sales
9	Executive
10	Finance
11	Accounting

```
select
department_name
from departments d
where
        EXISTS (select 1 from employees
                where department_id = d.department_id )
```

Subquery – correlation – NOT EXISTS

	department_name
1	Operations
2	Corporate Tax
3	Benefits
4	Payroll
5	Recruiting
6	Treasury
7	Retail Sales
8	IT Support

```
select
department_name
from departments d
where
        NOT EXISTS (select 1 from employees
                where department_id = d.department_id )
```

Problem

```
SQL Error [42703]: BŁĄD:
kolumna "full_salary" nie
istnieje
```

```sql
select
first_name,
last_name,
salary + (salary * coalesce(commission_pct,0)) full_salary
from employees e
where full_salary > 5000
```

Problem – resolve by subquery - table derrived

	first_name	last_name	full_salary
1	Steven	King	24,000
2	Neena	Yang	17,000
3	Lex	Garcia	17,000
4	Alexander	James	9,000
5	Bruce	Miller	6,000
6	Nancy	Gruenberg	12,008
7	Daniel	Faviet	9,000

```
select
        * from(
                select
                first_name,
                last_name,
                salary + (salary *
                    coalesce(commission_pct,0)) full_salary
                from employees e ) t1
where t1.full_salary > 5000
```

Subquery in select - derrived column

	first_name	last_name	dep_name
1	Steven	King	Executive
2	Neena	Yang	Executive
3	Lex	Garcia	Executive
4	Alexander	James	IT
5	Bruce	Miller	IT

```
select
first_name,
last_name,
        (select department_name from departments where
                department_id = e.department_id) dep_name
from employees e
```

Subquery many columns

	last_name	salary
1	King	24,000
2	James	9,000
3	Gruenberg	12,008
4	Li	11,000
5	Fripp	8,200
6	Singh	14,000
7	Whalen	4,400
8	Martinez	13,000
9	Jacobs	6,500
10	Brown	10,000
11	Higgins	12,008

```
select
e.last_name,
e.salary
from employees e
where
(department_id, salary) in
        (select department_id, max(salary) from employees
                group by department_id)
```